

Technical Document #838: Database Systems - Comprehensive Overview

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NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Database caching stores frequently accessed data in memory. Redis and Memcached are popular caching solutions.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- Database connection pooling reuses database connections to reduce overhead.
- Relational databases organize data into tables with rows and columns.
- Database monitoring tracks performance metrics and identifies bottlenecks.